21 April 1982

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NOR, DIGITAL, KEY CODE EMJ

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NOR gate module..

1.2 Classification. This specification covers two classes of modules.

Class I and class II as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.1139 failures per million operating hours for class I or 0.6743 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, failure rate for class I shall be used for reliability prediction (see 6.3).

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be EMJ.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/259-1 for class I and M28787/259-2 for class II, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{DD}) of 3.0 to 18.0 Vdc ±5%.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

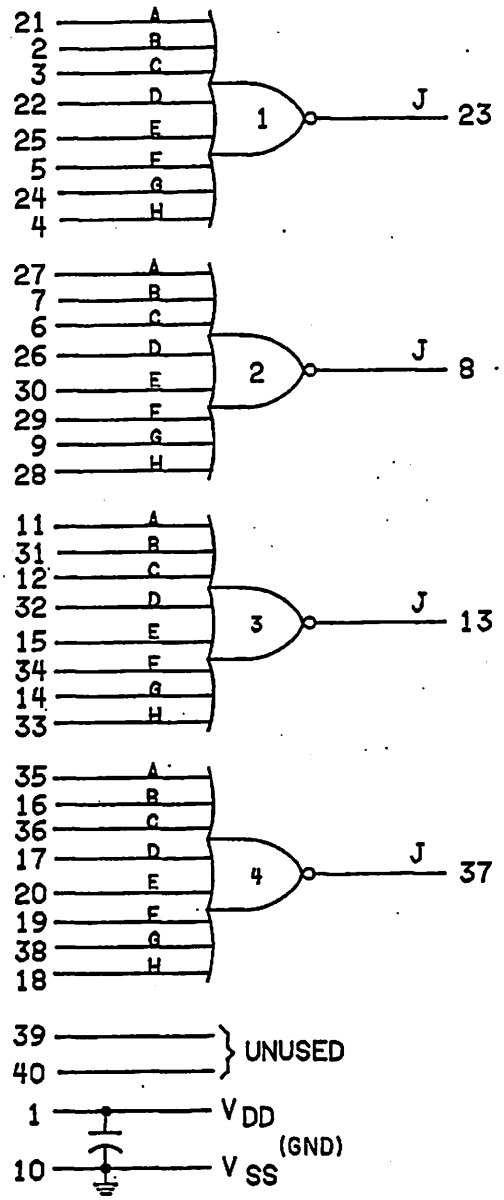


FIGURE 1. EMJ module function diagram, NOR gate.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1, A2	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B32	Input clamp diode voltage, negative	IV	X	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C32	Input clamp diode voltage, positive	V	X	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D4	Static output voltage, high-level, I_{OH} = 0	VI	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
E1-E4	Static output voltage, low-level, I_{OL} = 0	VI	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
F1-F4	Static output voltage, high level, V_{DD} = 5 V	VII	X	4.61	5.00	4.61	5.00	4.61	5.00	4.61	5.00	4.61	5.00	V
F5-F40	Static output voltage, low level, V_{DD} = 5 V	VII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G4	Static output voltage, high level, V_{DD} = 15 V	VIII	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V
G5-~~G6~~ G8	Static output voltage, low level, V_{DD} = 15 V	VIII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
H1-H32	Input current, high level	IX	X	0	200	---	---	0	200	---	---	0	200	nA
J1-J32	Input current, low level	X	X	0	-200	---	---	0	-200	---	---	0	-200	nA
K1	Static power supply current, low level	XI	X	0	8	0	8	0	80	0	8	0	160	µA
K2	Static power supply current, high level	XI	X	0	8	0	8	0	80	0	8	0	160	µA

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE 1. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L1-L4	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	XII	S	20	200	20	200	22	220	15	152	25	245	ns
L5-L8	Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	XII	Q	65	680	65	680	75	750	50	517	80	835	ns
M1-M4	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	XIII	S	20	200	20	200	22	220	15	152	25	245	ns
M5-M8	Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	XIII	Q	65	680	65	680	75	750	50	517	80	835	ns
N1-N4	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	XIV	Q	10	100	10	100	11	110	7	75	12	125	ns
N5-N8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	XIV	Q	10	100	10	100	11	110	7	75	12	125	ns
P1-P4	Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	XV	Q	8	80	8	80	9	90	6	60	10	100	ns
P5-P8	Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	XV	Q	8	80	8	80	9	90	6	60	10	100	ns
Q1-Q32	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	XVI	S	80	850	80	850	90	940	65	645	100	1040	ns
Q33-Q64	Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	XVI	Q	100	1050	100	1050	110	1160	80	800	120	1290	ns
R1-R32	Propagation delay time (t_{PHL}), C_L = 50 V, V_{DD} = 5 V	XVII	S	40	400	40	400	44	440	30	305	50	490	ns

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25° ±5°C 4/		Class I 4/				Class II 4/				Unit
						0° ±5°C		60° ±5°C		-55° ±5°C		100° ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
R33-R64	Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	XVII	Q	60	600	60	600	65	665	45	455	70	735	ns
S1-S32	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	XVIII	Q	30	340	30	340	35	375	25	260	40	415	ns
T1-T32	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	XIX	Q	15	160	15	160	17	175	12	120	19	195	ns
U1-U32	Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	XX	Q	24	240	24	240	26	264	18	185	30	295	ns
V1-V32	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	XXI	Q	12	120	12	120	13	135	9	90	15	145	ns
W	Power supply filter capacitance	XXII	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
X1-X32	Input capacitance	XXIII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

4.4.2 Order of testing. Test conditions A through X shall be performed in any order with the following exceptions. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, C where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through X in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B through G and K through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. Test conditions H and J shall be performed at 25°, 60° and 100°C only using the initial limits for the applicable temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Transient thermal test. Use the same test conditions as the maximum class temperature conditions.

4.5.1.2 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition W, shall be performed using the initial limits at 25° ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests B through W shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests B through X shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

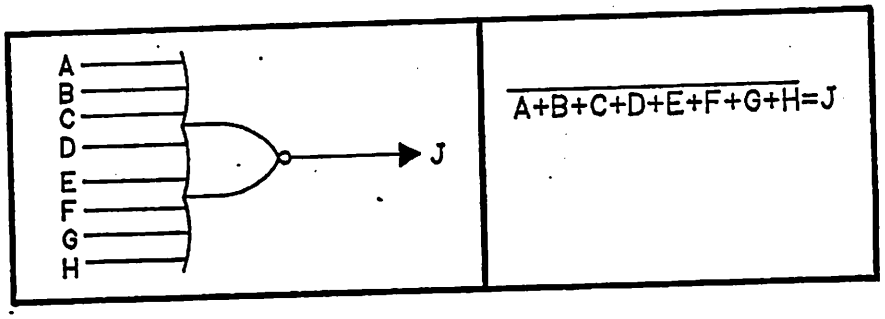
6.4 Module description. The following electrical parameters reflect the maximum class temperature end-of-life limits from table I.

	Class I 2.4	Class II 2.4	Units
Maximum power dissipation			mW
Input voltage			
Low level (V_{DD} = 5 V)	0 min, 1.5 max	0 min, 1.5 max	V
(V_{DD} = 15 V)	0 min, 4 max	0 min, 4 max	V
High level (V_{DD} = 5 V)	3.5 min, V_{DD} max	3.5 min, V_{DD} max	V
(V_{DD} = 15 V)	11 min, V_{DD} max	11 min, V_{DD} max	V
Input current			
Low level (V_{IL} = 0.0 V) (V_{DD} = 18 V)	0 min, -200 max	0 min, -200 max	nA
High level (V_{IH} = 18 V) (V_{DD} = 18 V)	0 min, 200 max	0 min, 200 max	nA
Output current			
Low level (V_{OL} = 0.4 V) (V_{DD} = 5 V)			
I_{OL} at -55°C	---	0.64 max	mA
0° and 25°C	0.51 max	0.51 max	mA
60°C	0.44 max	0.44 max	mA
100°C	---	0.36 max	mA
High level (V_{OH} = 4.6 V) (V_{DD} = 5 V)			
I_{OH} at -55°C	---	-0.64 max	mA
0° and 25°C	-0.51 max	-0.51 max	mA
60°C	-0.44 max	-0.44 max	mA
100°C	---	-0.36 max	mA

6.5 Module function. The module contains four independent 8-input NOR gates.

6.5.1 Input to output relationship. Each gate shall have the input to output relationship of table II for positive logic.

TABLE II. Input to output relationship for positive logic.



Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

6.5.2 Dynamic power dissipation. The dynamic power dissipation can be approximated by the formula:

$$P_{TOTAL} = (C\ V_{DD}^2 f)\ N + \text{Quiescent power dissipation.}$$

Where: C = load capacitance.
V_{DD} = Supply voltage.
f = operating frequency.
N = Number of outputs.

6.6 Module handling considerations. To avoid possible damage to the devices during handling, testing, or actual operation, the following procedures should be followed:

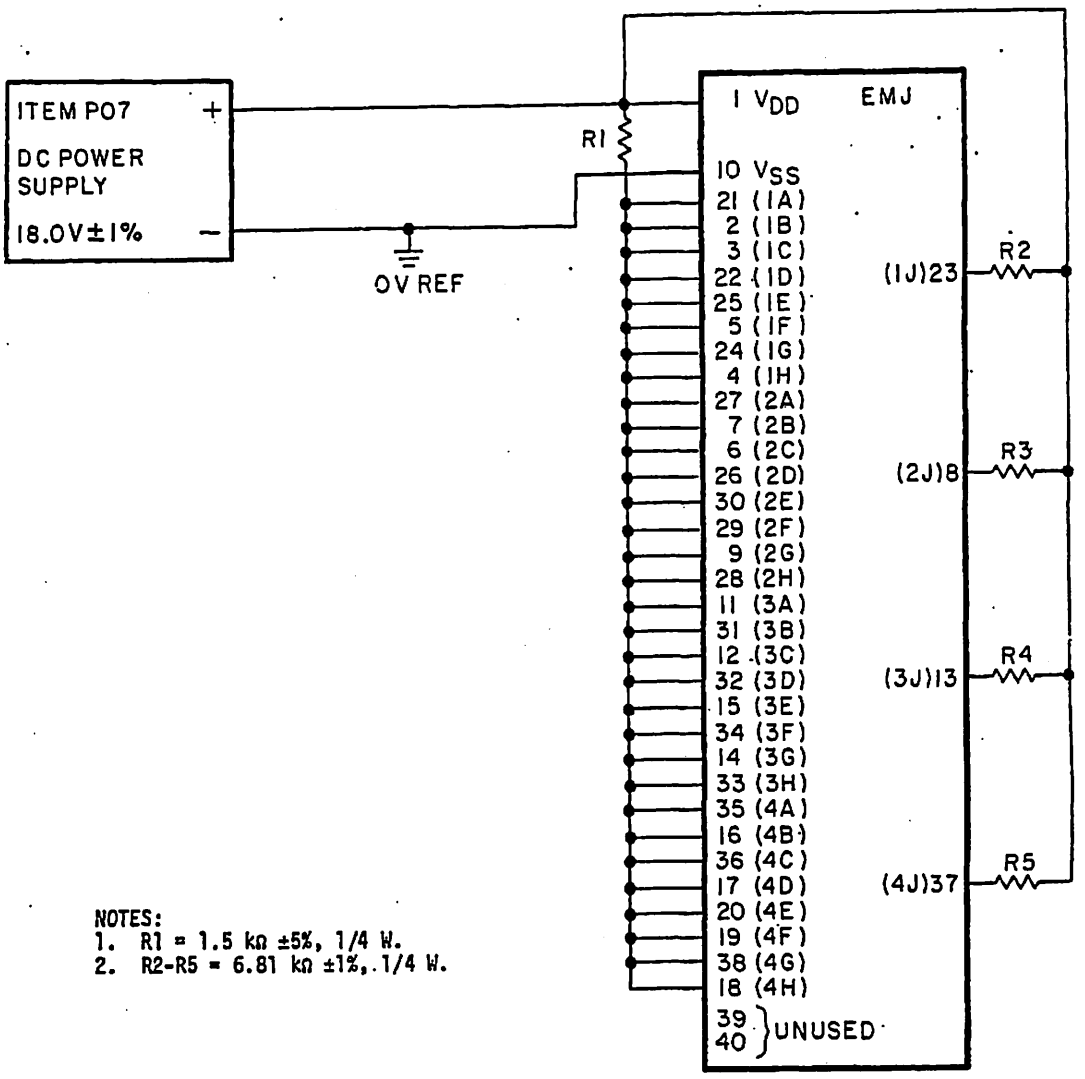
a. Device pins should be in contact with conductive material except when being tested or in actual operation, to avoid build-up of static charge.
b. Soldering iron tips, metal parts of fixtures and tools, and handling facilities should be grounded.
c. Signals should not be applied to the inputs while the device power supply is off.
d. All input signal and condition voltages must be removed before V_{DD} is disconnected.
e. Input voltages greater than the supply (V_{DD} - V_{SS}) should not be applied as this will forward bias the protection input diodes.
f. All unused input leads must be connected to either V_{SS} (GND) or V_{DD} (device supply), whichever is appropriate for the logic circuit involved.

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5963-N256)

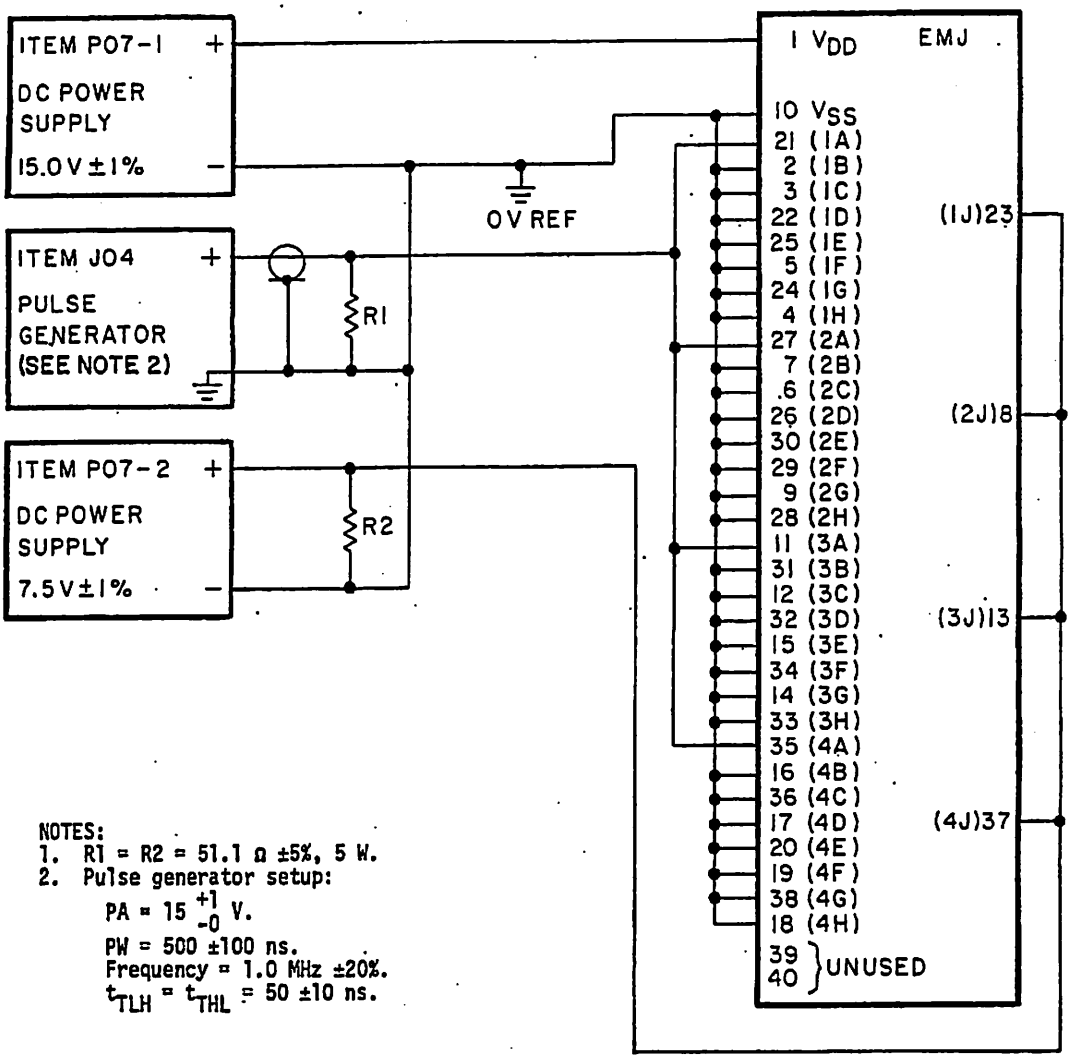
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 1.5 kΩ ±5%, 1/4 W.
2. R2-R5 = 6.81 kΩ ±1%, 1/4 W.

FIGURE 2. Operating temperature and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1 Ω ±5%, 5 W.
2. Pulse generator setup:
   PA = 15 $^{+1}_{-0}$ V.
   PW = 500 ±100 ns.
   Frequency = 1.0 MHz ±20%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.

FIGURE 3. Component temperature test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

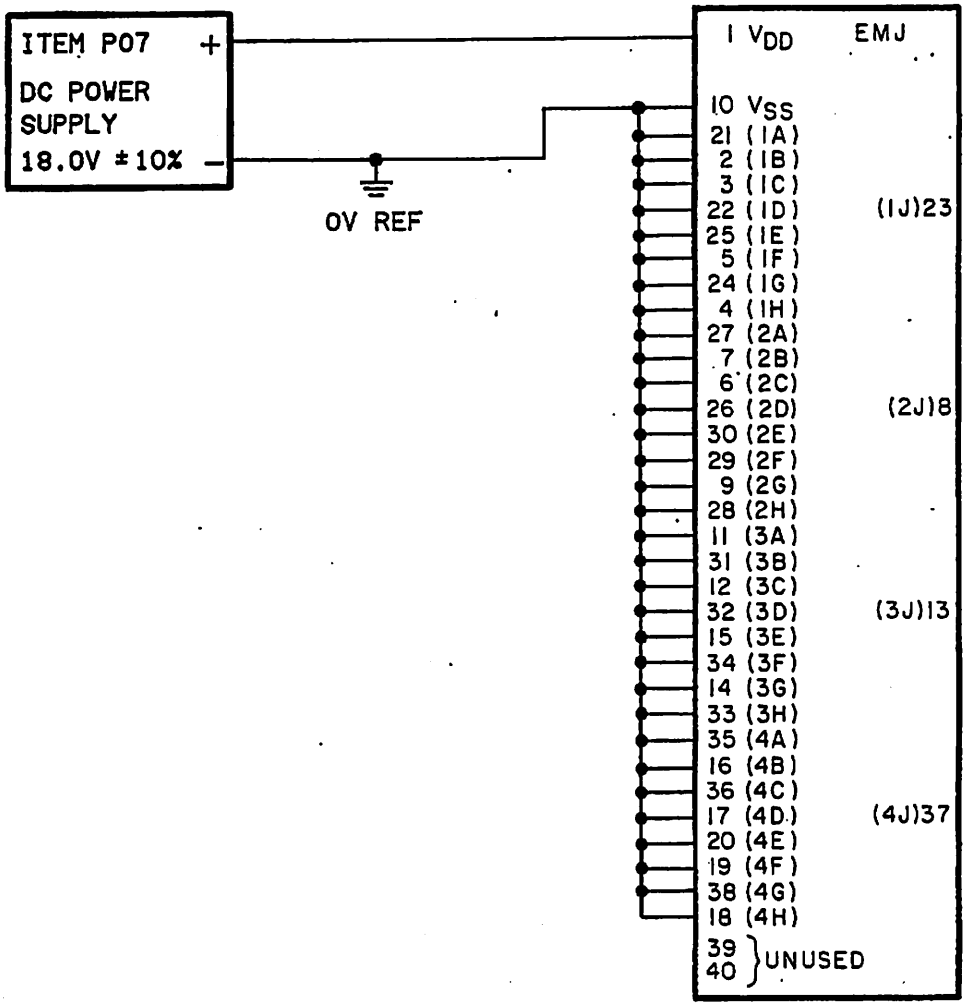


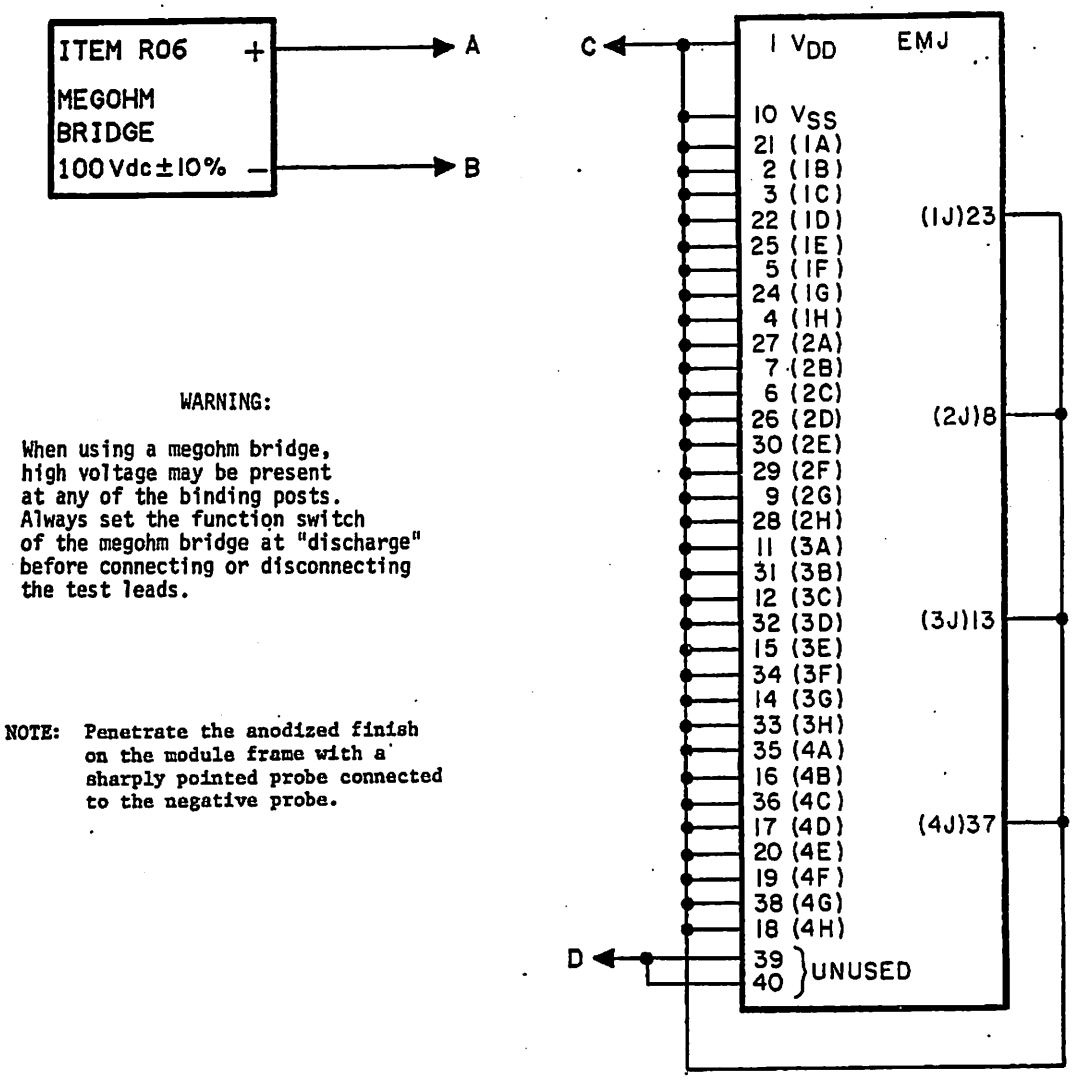
FIGURE 4. Vibration test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C, D	Frame	10	∞	MΩ	5
A2	Isolation	C	D	10	∞	MΩ	5

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



WARNING:

When using a megohm bridge, high voltage may be present at any of the binding posts. Always set the function switch of the megohm bridge at "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative probe.

FIGURE 5. Isolation test circuit for table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE IV. Input clamp diode voltage (negative) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B1	1A Input clamp diode voltage, negative	21	Open	-0.4	-6.0	V	6
B2	1B	2					
B3	1C	3					
B4	1D	22					
B5	1E	25					
B6	1F	5					
B7	1G	24					
B8	1H	4					
B9	2A	27					
B10	2B	7					
B11	2C	6					
B12	2D	26					
B13	2E	30					
B14	2F	29					
B15	2G	9					
B16	2H	28					
B17	3A	11					
B18	3B	31					
B19	3C	12					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE IV. Input clamp diode voltage (negative) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
B20	3D Input clamp diode voltage, negative	32	Open	-0.4	-6.0	V	6
B21	3E	15					
B22	3F	34					
B23	3G	14					
B24	3H	33					
B25	4A	35					
B26	4B	16					
B27	4C	36					
B28	4D	17					
B29	4E	20					
B30	4F	19					
B31	4G	38					
B32	4H	18					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage (positive) test.

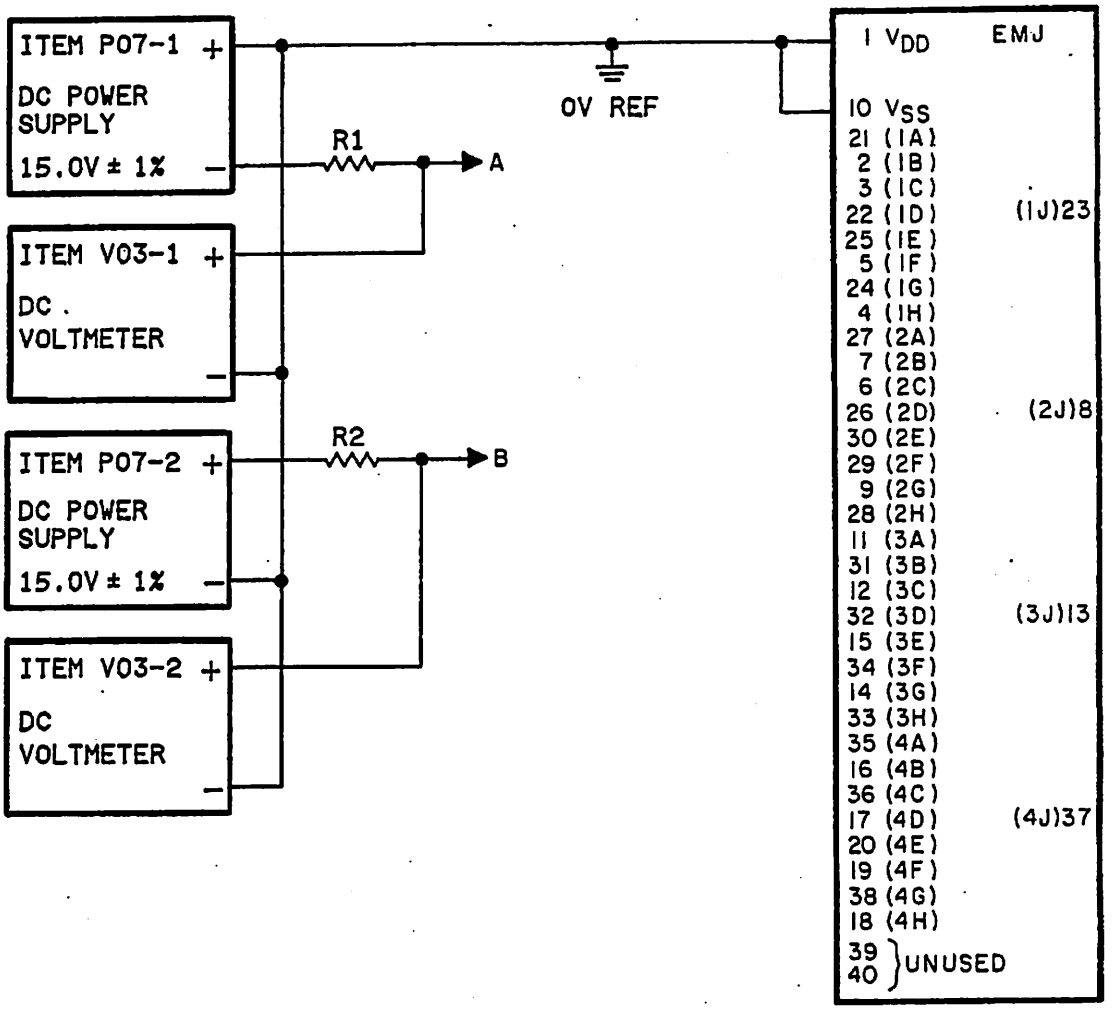
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C1	1A Input clamp diode voltage, positive	Open	21	0.4	1.5	V	6
C2	1B		2				
C3	1C		3				
C4	1D		22				
C5	1E		25				
C6	1F		5				
C7	1G		24				
C8	1H		4				
C9	2A		27				
C10	2B		7				
C11	2C		6				
C12	2D		26				
C13	2E		30				
C14	2F		29				
C15	2G		9				
C16	2H		28				
C17	3A		11				
C18	3B		31				
C19	3C		12				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE V. Input clamp diode voltage (positive) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C20	3D Input clamp diode voltage, positive	Open	32	0.4	1.5	V	6
C21	3E		15				
C22	3F		34				
C23	3G		14				
C24	3H		33				
C25	4A		35				
C26	4B		16				
C27	4C		36				
C28	4D		17				
C29	4E		20				
C30	4F		19				
C31	4G		38				
C32	4H		18				

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:10Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 9.09 kΩ ±1%, 1/8 W.
2. R2 = 13.7 kΩ ±1%, 1/8 W.
3. Since dc power supplies P07-1 and P07-2 and voltmeters V03-1 and V03-2 are not used simultaneously, only one dc power supply P07 and one voltmeter V03 is required.

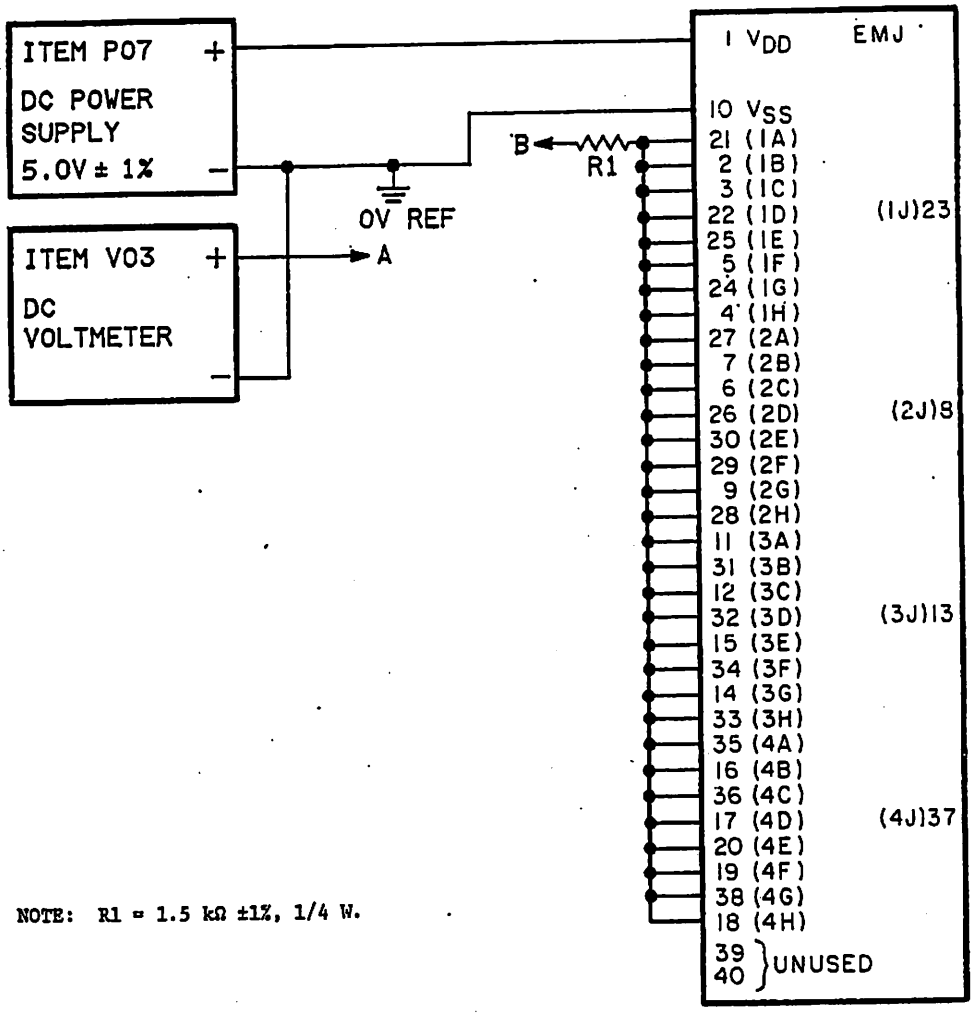
FIGURE 6. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE VI. Static output voltage (high-level and low-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
D1	1J Static output voltage, high level, I_{OH} = 0	23	10	4.95	5.05	V	7
D2	2J	8					
D3	3J	13					
D4	4J	37					
E1	1J Static output voltage, low level, I_{OL} = 0	23	1	0.00	0.05		
E2	2J	8					
E3	3J	13					
E4	4J	37					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



NOTE: R1 = 1.5 kΩ ±1%, 1/4 W.

FIGURE 7. Static output voltage (high-level and low-level) test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE VII. Static output voltage (high-level and low-level, V_{DD} = 5 V) test.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F1	1J Static output voltage, high level, V_{DD} = 5 V	2,3,4,5,6,7, 9,11,12,14, 15,16,17,18, 19,20,21,22, 24,25,26,27, 28,29,30,31, 32,33,34,35, 36,38	Open	23	Open	4.61	5.00	V	8
F2	2J			8					
F3	3J			13					
F4	4J			37					
F5	1J Static output voltage, low level, V_{DD} = 5 V	2,3,4,5,6,7, 9,12,14,15, 16,17,18,19, 20,22,24,25, 26,28,29,30, 31,32,33,34, 36,38	11,21,27,35	Open	23	0.0	0.4		
F6	2J				8				
F7	3J				13				
F8	4J				37				
F9	1J	3,4,5,6,9, 11,12,14,15, 17,18,19,20, 21,22,24,25, 26,27,28,29, 30,32,33,34, 35,36,38	2,7,16,31		23				
F10	2J				8				
F11	3J				13				
F12	4J				37				
F13	1J	2,4,5,7,9, 11,14,15,16, 17,18,19,20, 21,22,24,25, 26,27,28,29, 30,31,32,33, 34,35,38	3,6,12,36		23				
F14	2J				8				
F15	3J				13				
F16	4J				37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE VII. Static output voltage (high-level and low-level, V_{DD} = 5 V) test - Continued.

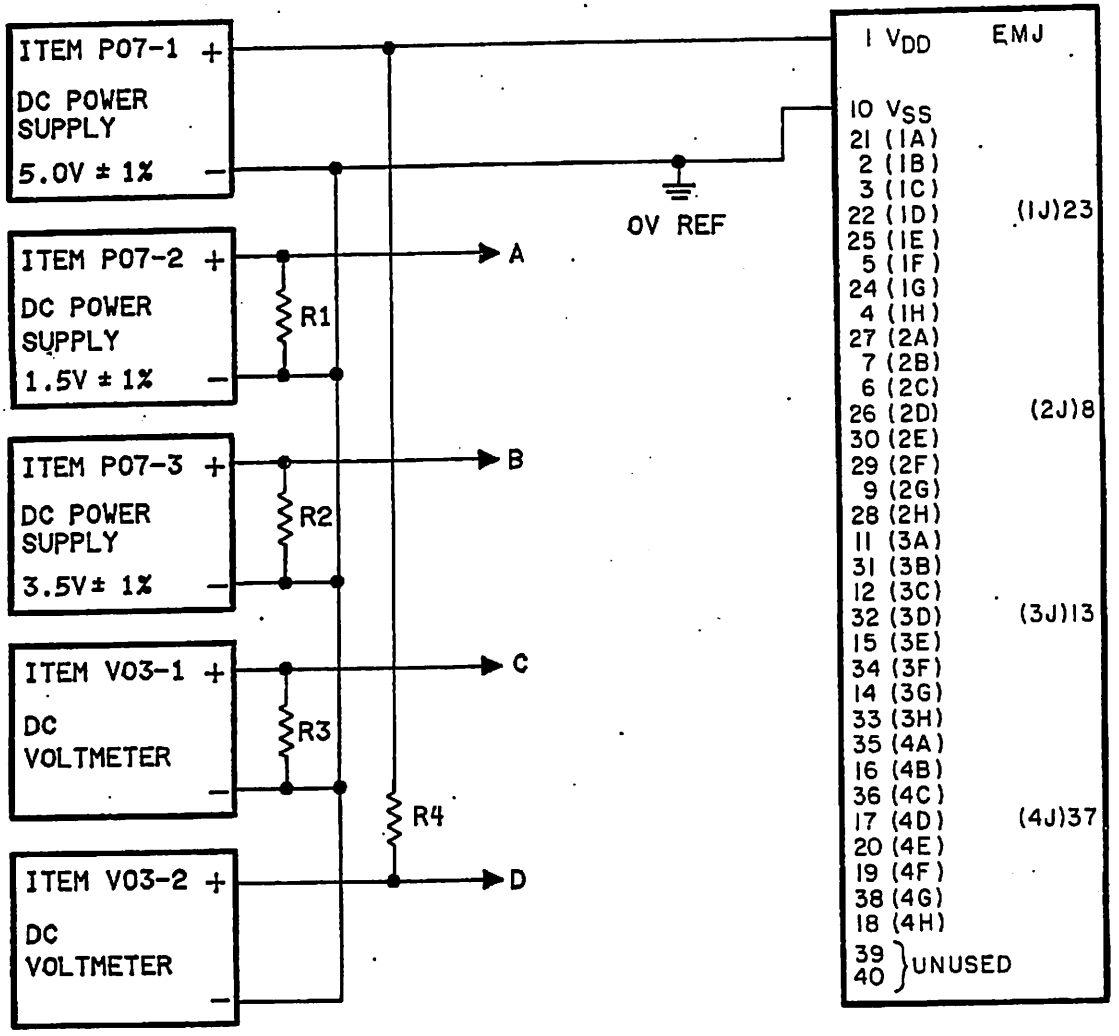
Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F17	1J Static output voltage, low level, V_{DD} = 5 V	2,3,4,5,6,7, 9,11,12,14, 15,16,18,19, 20,21,24,25, 27,28,29,30, 31,33,34,35, 36,38	17,22,26,32	Open	23	0.0	0.4	V	8
F18	2J				8				
F19	3J				13				
F20	4J				37				
F21	1J	2,3,4,5,6,7, 9,11,12,14, 16,17,18,19, 21,22,24,26, 27,28,29,31, 32,33,34,35, 36,38	15,20,25,30		23				
F22	2J				8				
F23	3J				13				
F24	4J				37				
F25	1J	2,3,4,6,7,9, 11,12,14,15, 16,17,18,20, 21,22,24,25, 26,27,28,30, 31,32,33,35, 36,38	5,19,29,34		23				
F26	2J				8				
F27	3J				13				
F28	4J				37				
F29	1J	2,3,4,5,6,7, 11,12,15,16, 17,18,19,20, 21,22,25,26, 27,28,29,30, 31,32,33,34, 35,36	9,14,24,38		23				
F30	2J				8				
F31	3J				13				
F32	4J				37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE VII. Static output voltage (high-level and low-level, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect				Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
F33	1J Static output voltage, low level, V_{DD} = 5 V	2,3,5,6,7,9, 11,12,14,15, 16,17,19,20, 21,22,24,25, 26,27,29,30, 31,32,34,35, 36,38	4,18,28,33	Open	23	0.0	0.4	V	8
F34	2J				8				
F35	3J				13				
F36	4J				37				
F37	1J	Open	2,3,4,5,6,7, 9,11,12,14, 15,16,17,18, 19,20,21,22, 24,25,26,27, 28,29,30,31, 32,33,34,35, 36,38		23				
F38	2J				8				
F39	3J				13				
F40	4J				37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = R2 = 51.1 Ω ±5%.
2. R3 = 18.2 kΩ ±1%, 1/4 W @ −55°C.
   = 23.2 kΩ ±1%, 1/4 W @ 0° and 25°C.
   = 27.4 kΩ ±1%, 1/4 W @ 60°C.
   = 32.4 kΩ ±1%, 1/4 W @ 100°C.
3. R4 = 7.15 kΩ ±1%, 1/4 W @ −55°C.
   = 9.09 kΩ ±1%, 1/4 W @ 0° and 25°C.
   = 10.5 kΩ ±1%, 1/4 W @ 60°C.
   = 12.7 kΩ ±1%, 1/4 W @ 100°C.

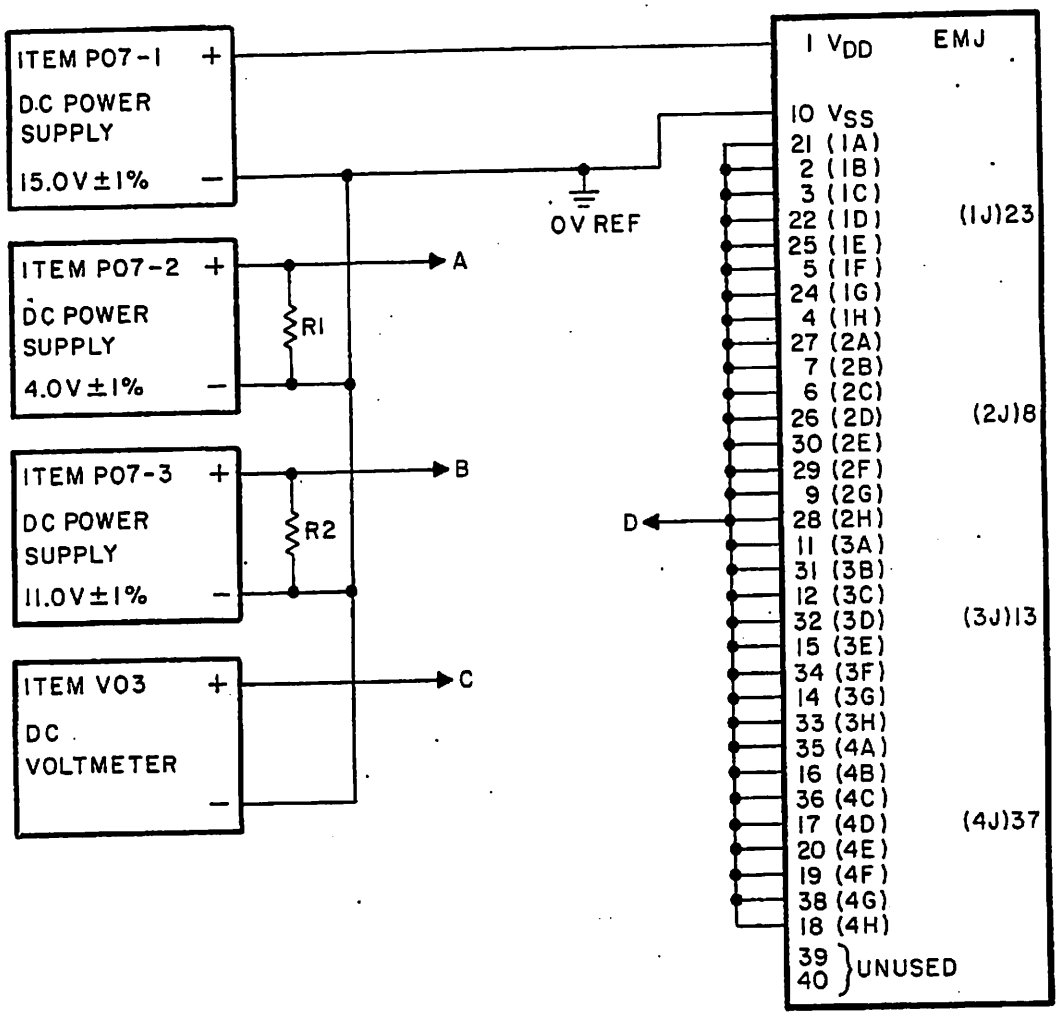
FIGURE 8. Static output voltage (high-level and low-level, V_{DD} = 5 V) test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE VIII. Static output voltage (high-level and low-level, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1J Static output voltage, high level, V_{DD} = 15 V	D	Open	23	13.5	15.0	V	9
G2	2J			8				
G3	3J			13				
G4	4J			37				
G5	1J Static output voltage, low level, V_{DD} = 15 V	Open	D	23	0.0	1.5		
G6	2J			8				
G7	3J			13				
G8	4J			37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



NOTES: 1. R1 = 51.1 Ω ±5%, 5W.
2. R2 = 576 Ω ±5%, 1/4 W.

FIGURE 9. Static output voltage (high-level and low-level, V_{DD} = 15V) test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H1	1A Input current, high level	2,3,22,25,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	21	0	200	nA	10
H2	1B	21,3,22,25,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	2				
H3	1C	21,2,22,25,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	3				
H4	1D	21,2,3,25,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	22				
H5	1E	21,2,3,22,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	25				
H6	1F	21,2,3,22,25,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	5				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H7	1G Input current, high level	21,2,3,22,25,5, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	24	0	200	nA	10
H8	1H	21,2,3,22,25,5, 24,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	4				
H9	2A	21,2,3,22,25,5, 24,4,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	27				
H10	2B	21,2,3,22,25,5, 24,4,27,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	7				
H11	2C	21,2,3,22,25,5, 24,4,27,7,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	6				
H12	2D	21,2,3,32,25,5, 24,4,27,7,6,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	26				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H13	2E Input current, high level	21,2,3,22,25,5, 24,4,27,7,6,26, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	30	0	200	nA	10
H14	2F	21,2,3,22,25,5, 24,4,27,7,6,26, 30,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	29				
H15	2G	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	9				
H16	2H	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	28				
H17	3A	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	11				
H18	3B	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	31				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
H19	3C Input current, high level	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,32,15,34,14, 33,35,16,36,17, 20,19,38,18	12	0	200	nA	10
H20	3D	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,15,34,14, 33,35,16,36,17, 20,19,38,18	32				
H21	3E	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,34,14, 33,35,16,36,17, 20,19,38,18	15				
H22	3F	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,14, 33,35,16,36,17, 20,19,38,18	34				
H23	3G	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 33,35,16,36,17, 20,19,38,18	14				
H24	3H	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,35,16,36,17, 20,19,38,18	33				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H25	4A Input current, high level	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,16,36,17, 20,19,38,18	35	0	200	nA	10
H26	4B	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,36,17, 20,19,38,18	16				
H27	4C	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,17, 20,19,38,18	36				
H28	4D	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 20,19,38,18	17				
H29	4E	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,19,38,18	20				
H30	4F	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,20,38,18	19				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE IX. Input current (high-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
H31	4G Input current, high level	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,20,19,18	38	0	200	nA	10
H32	4H Input current, high level	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,20,19,38	18	0	200	nA	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

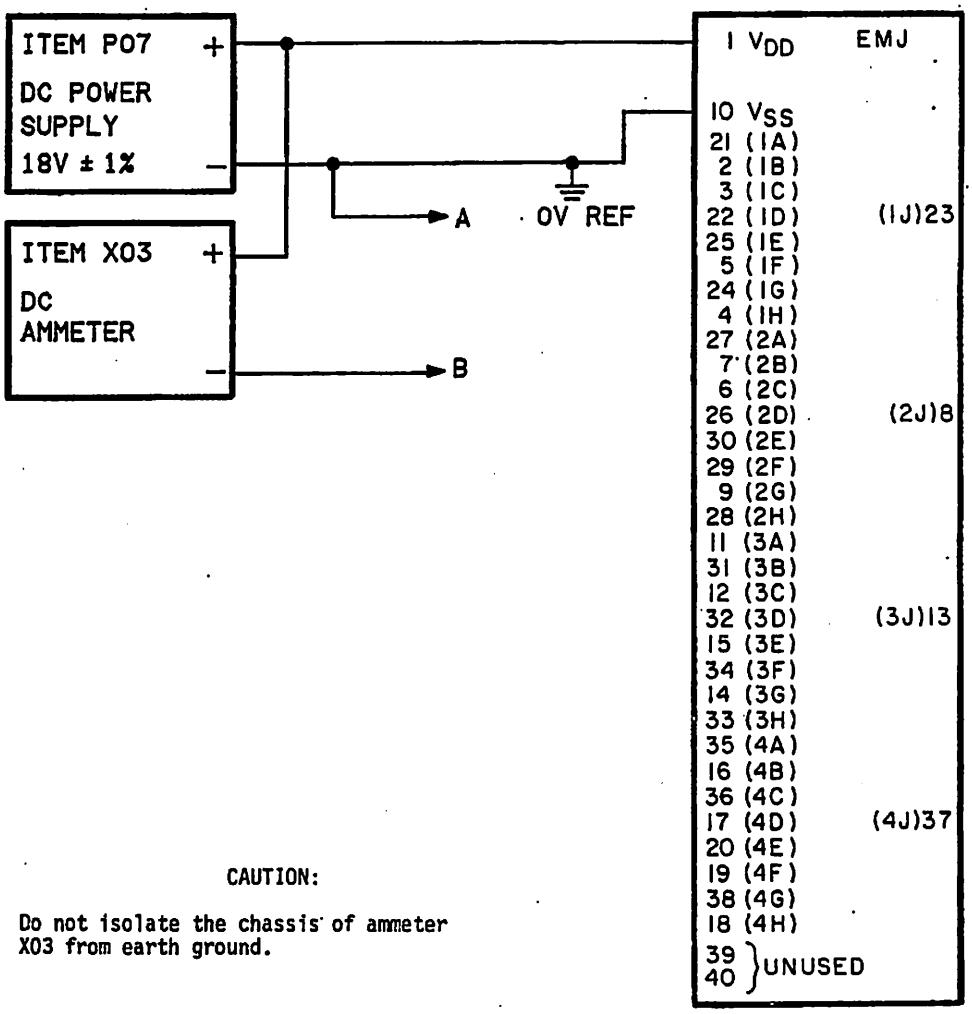


FIGURE 10. Input current (high-level) test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE X. Input current (low-level) test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J1	1A Input current, low level	2,3,22,25,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	21	0	-200	nA	11
J2	1B	21,3,22,25,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	2				
J3	1C	21,2,22,25,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	3				
J4	1D	21,2,3,25,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	22				
J5	1E	21,2,3,22,5,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	25				
J6	1F	21,2,3,22,25,24, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	5				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE X. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J7	1G Input current, low level	21,2,3,22,25,5, 4,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	24	0	-200	nA	11
J8	1H	21,2,3,22,25,5, 24,27,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	4				
J9	2A	21,2,3,22,25,5, 24,4,7,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	27				
J10	2B	21,2,3,22,25,5, 24,4,27,6,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	7				
J11	2C	21,2,3,22,25,5, 24,4,27,7,26,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	6				
J12	2D	21,2,3,22,25,5, 24,4,27,7,6,30, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	26				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE X. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J13	2E Input current, low level	21,2,3,22,25,5, 24,4,27,7,6,26, 29,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	30	0	-200	nA	11
J14	2F	21,2,3,22,25,5, 24,4,27,7,6,26, 30,9,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	29				
J15	2G	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,28,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	9				
J16	2H	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,11,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	28				
J17	3A	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,31, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	11				
J18	3B	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 12,32,15,34,14, 33,35,16,36,17, 20,19,38,18	31				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE X. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J19	3C Input current, low level	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,32,15,34,14, 33,35,16,36,17, 20,19,38,18	12	0	-200	nA	11
J20	3D	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,15,34,14, 33,35,16,36,17, 20,19,38,18	32				
J21	3E	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,34,14, 33,35,16,36,17, 20,19,38,18	15				
J22	3F	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,14, 33,35,16,36,17, 20,19,38,18	34				
J23	3G	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 33,35,16,36,17, 20,19,38,18	14				
J24	3H	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,35,16,36,17, 20,19,38,18	33				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE X. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J25	4A Input current, low level	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,16,36,17, 20,19,38,18	35	0	-200	nA	11
J26	4B	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,36,17, 20,19,38,18	16				
J27	4C	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,17, 20,19,38,18	36				
J28	4D	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 20,19,38,18	17				
J29	4E	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,19,38,18	20				
J30	4F	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,20,38,18	19				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

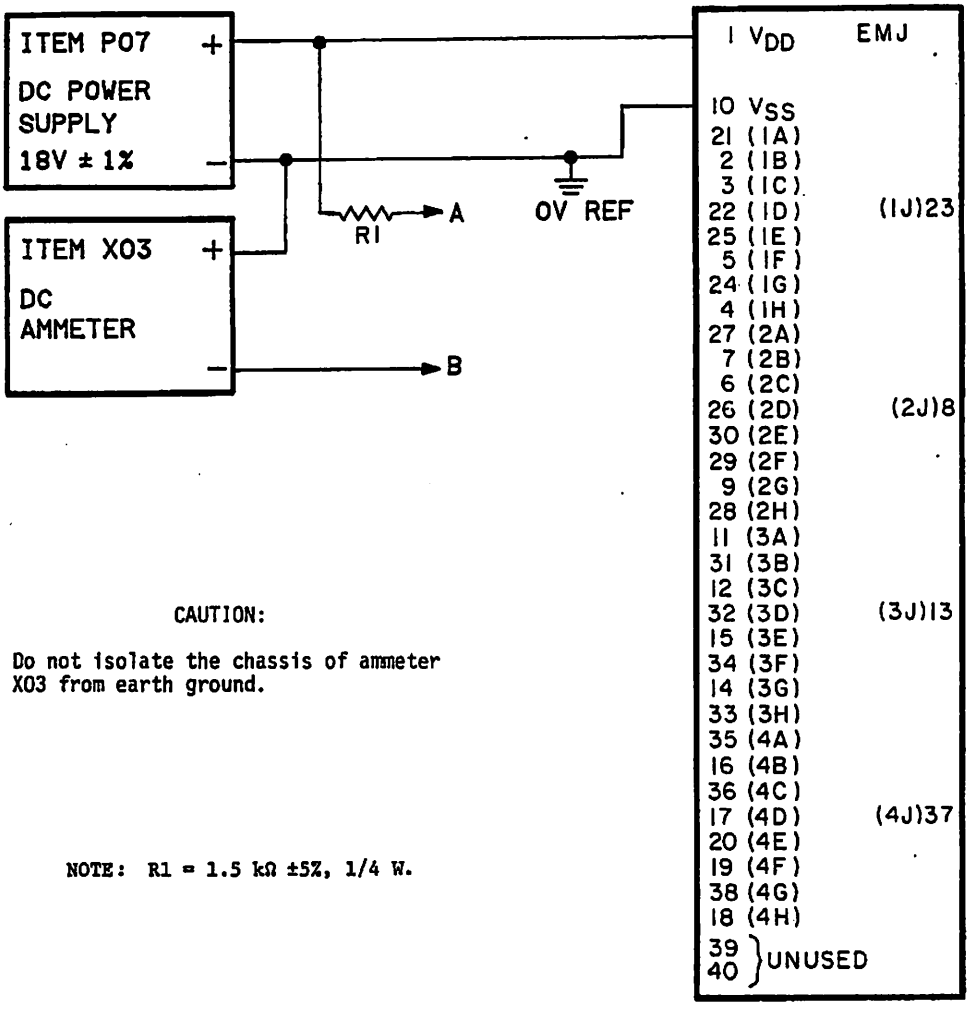
TABLE X. Input current (low-level) test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J31	4G Input current, low level	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,20,19,18	38	0	-200	nA	11
J32	4H Input current, low level	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,20,19,38	18	0	-200	nA	11

TABLE XI. Static power supply current (low-level and high-level) test.

Test condition	Parameter	Connect		Initial limits 25° +5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
K1	Static power supply current, low level	Open	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,20,19,38,18	0	8	µA	12
K2	Static power supply current, high level	21,2,3,22,25,5, 24,4,27,7,6,26, 30,29,9,28,11, 31,12,32,15,34, 14,33,35,16,36, 17,20,19,38,18	Open	0	8	µA	12

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T17:10Z
Check the source to verify that this is the current version before use.



CAUTION:

Do not isolate the chassis of ammeter X03 from earth ground.

NOTE: R1 = 1.5 kΩ ±5%, 1/4 W.

FIGURE 11. Input current (low-level) test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

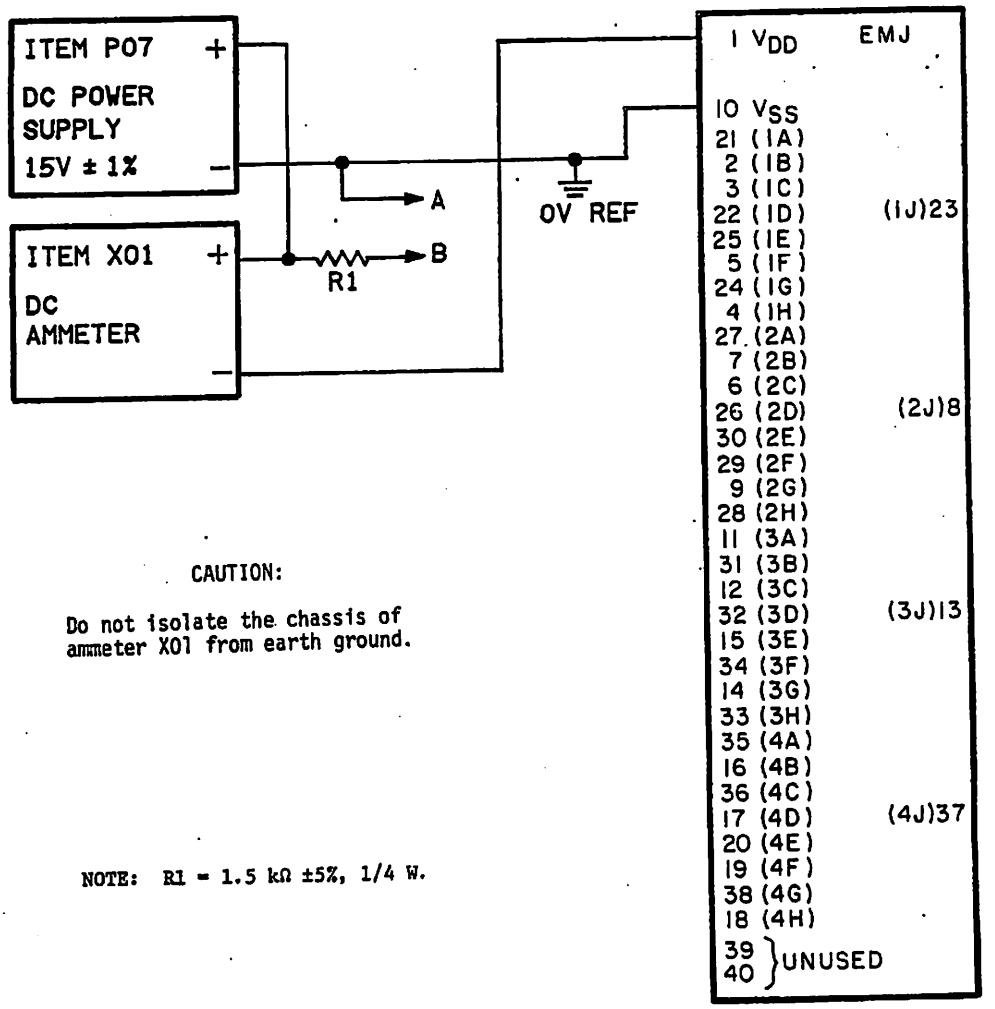


FIGURE 12. Static power supply current (low-level and high-level) test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XII. Transition time (t_{TLH}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
L1	1J Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 5 V	27,11,35	21	23	20	200	ns	13,15
L2	2J	21,11,35	27	8				
L3	3J	21,27,35	11	13				
L4	4J	21,27,11	35	37				
L5	1J Transition time (t_{TLH}), C_L = 150 pF, V_{DD} = 5 V	27,11,35	21	23	65	680		
L6	2J	21,11,35	27	8				
L7	3J	21,27,35	11	13				
L8	4J	21,27,11	35	37				

TABLE XIII. Transition time (t_{THL}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
M1	1J Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 5 V	27,11,35	21	23	20	200	ns	13,15
M2	2J	21,11,35	27	8				
M3	3J	21,27,35	11	13				
M4	4J	21,27,11	35	37				
M5	1J Transition time (t_{THL}), C_L = 150 pF, V_{DD} = 5 V	27,11,35	21	23	65	680		
M6	2J	21,11,35	27	8				
M7	3J	21,27,35	11	13				
M8	4J	21,27,11	35	37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

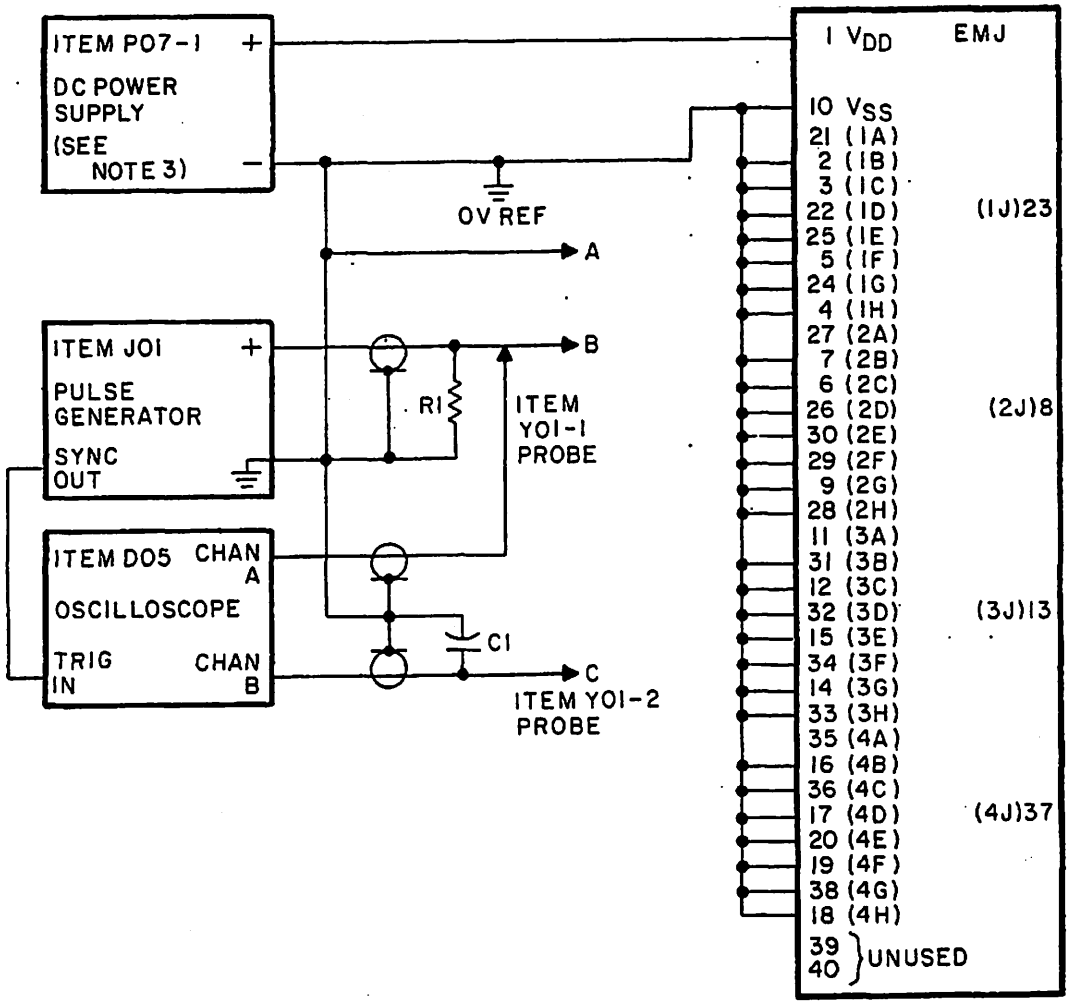
TABLE XIV. Transition time (t_{TLH} and t_{THL}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1J Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 10 V	27,11,35	21	23	10	100	ns	13,15
N2	2J	21,11,35	27	8				
N3	3J	21,27,35	11	13				
N4	4J	21,27,11	35	37				
N5	1J Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 10 V	27,11,35	21	23				
N6	2J	21,11,35	27	8				
N7	3J	21,27,35	11	13				
N8	4J	21,27,11	35	37				

TABLE XV. Transition time (t_{TLH} and t_{THL}, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
P1	1J Transition time (t_{TLH}), C_L = 50 pF, V_{DD} = 15 V	27,11,35	21	23	8	80	ns	13,15
P2	2J	21,11,35	27	8				
P3	3J	21,27,35	11	13				
P4	4J	21,27,11	35	37				
P5	1J Transition time (t_{THL}), C_L = 50 pF, V_{DD} = 15 V	27,11,35	21	23				
P6	2J	21,11,35	27	8				
P7	3J	21,27,35	11	13				
P8	4J	21,27,11	35	37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



NOTES:
1. R1 = 51.1 Ω ±5%, 1/2 W.
2. C1 = 50 ±5 pF for test conditions L1-L4, M1-M4, N1-N8 and P1-P8, including probe and parasitic capacitance.
   = 150 ±15 pF for test conditions L5-L8 and M5-M8, including probe and parasitic capacitance.
3. Item P07-1 Voltage:
   = 5.0 V ±1% for test conditions L1-L8 and M1-M8.
   = 10.0 V ±1% for test conditions N1-N8.
   = 15.0 V ±1% for test conditions P1-P8.

FIGURE 13. Transition time (t_{TLH} and t_{THL}) test circuit for tables XII, XIII, XIV and XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q1	1A to 1J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	2,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	21	23	80	850	ns	14,15
Q2	1B to 1J	21,3,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	2					
Q3	1C to 1J	21,2,22,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	3					
Q4	1D to 1J	21,2,3,25,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	22					
Q5	1E to 1J	21,2,3,22,5,24,4,27,7,6,26,30,29,9,28,11,31,12,32,15,34,14,33,35,16,36,17,20,19,38,18	25					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q6	1F to 1J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	5	23	80	850	ns	14,15
Q7	1G to 1J	21,2,3,22, 25,5,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	24					
Q8	1H to 1J	21,2,3,22, 25,5,24,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	4					
Q9	2A to 2J	21,2,3,22, 25,5,24,4,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	27	8				
Q10	2B to 2J	21,2,3,22, 25,5,24,4, 27,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	7	8				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin.	Min	Max		
Q11	2C to 2J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	6	8	80	850	ns	14,15
Q12	2D to 2J	21,2,3,22, 25,5,24,4, 27,7,6,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	26					
Q13	2E to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	30					
Q14	2F to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	29					
Q15	2G to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	9					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
Q16	2H to 2J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	28	8	80	850	ns	14,15
Q17	3A to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	11	13				
Q18	3B to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	31					
Q19	3C to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	12					
Q20	3D to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,15, 34,14,33,35, 16,36,17,20, 19,38,18	32					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q21	3E to 3J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 34,14,33,35, 16,36,17,20, 19,38,18	15	13	80	850	ns	14,15
Q22	3F to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,14,33,35, 16,36,17,20, 19,38,18	34					
Q23	3G to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,33,35, 16,36,17,20, 19,38,18	14					
Q24	3H to 3G	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,35, 16,36,17,20, 19,38,18	33					
Q25	4A to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 16,36,17,20, 19,38,18	35	37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q26	4B to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,36,17,20, 19,38,18	16	37	80	850	ns	14,15
Q27	4C to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,17,20, 19,38,18	36					
Q28	4D to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,20, 19,38,18	17					
Q29	4E to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 19,38,18	20					
Q30	4F to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,38,18	19					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q31	4G to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,18	38	37	80	850	ns	14,15
Q32	4H to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,38	18	37	80	850		
Q33	1A to 1J Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	2,3,22,25,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	21	23	100	1050		
Q34	1B to 1J	21,3,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	2					
Q35	1C to 1J	21,2,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	3					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q36	1D to 1J Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,25, 5,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	22	23	100	1050	ns	14,15
Q37	1E to 1J	21,2,3,22,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	25					
Q38	1F to 1J	21,2,3,22, 25,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	5					
Q39	1G to 1J	21,2,3,22, 25,5,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	24					
Q40	1H to 1J	21,2,3,22, 25,5,24,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	4					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q41	2A to 2J Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	27	8	100	1050	ns	14,15
Q42	2B to 2J	21,2,3,22, 25,5,24,4, 27,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	7					
Q43	2C to 2J	21,2,3,22, 25,5,24,4, 27,7,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	6					
Q44	2D to 2J	21,2,3,22, 25,5,24,4, 27,7,6,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	26					
Q45	2E to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	30					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
Q46	2F to 2J Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	29	8	100	1050	ns	14,15
Q47	2G to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	9					
Q48	2H to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29, 9,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	28					
Q49	3A to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 31,12,34,15, 32,14,33,35, 16,36,17,20, 19,38,18	11	13				
Q50	3B to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	31	13				

Source: https://assist.dla.mil-Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q51	3C to 3J Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	12	13	100	1050	ns	14,15
Q52	3D to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,15, 34,14,33,35, 16,36,17,20, 19,38,18	32					
Q53	3E to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 34,14,33,35, 16,36,17,20, 19,38,18	15					
Q54	3F to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,14,33,35, 16,36,17,20, 19,38,18	34					
Q55	3G to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,33,35, 16,36,17,20, 19,38,18	14					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q56	3H to 3G Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,35, 16,36,17,20, 19,38,18	33	13	100	1050	ns	14,15
Q57	4A to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 16,36,17,20, 19,38,18	35	37				
Q58	4B to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,36,17,20, 19,38,18	16					
Q59	4C to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,17,20, 19,38,18	36					
Q60	4D to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,20, 19,38,18	17					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVI. Propagation delay time (t_{PLH}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
Q61	4E to 4J Propagation delay time (t_{PLH}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 19,38,18	20	37	100	1050	ns	14,15
Q62	4F to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,38,18	19					
Q63	4G to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,18	38					
Q64	4H to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,38	18					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R1	1A to 1J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	2,3,22,25, 5,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	21	23	40	400	ns	14,15
R2	1B to 1J	21,3,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	2					
R3	1C to 1J	21,2,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	3					
R4	1D to 1J	21,2,3,25,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	22					
R5	1E to 1J	21,2,3,22,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	25					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
R6	1F to 1J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	5	23	40	400	ns	14,15
R7	1G to 1J	21,2,3,22, 25,5,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	24	↓	↓	↓	↓	↓
R8	1H to 1J	21,2,3,22, 25,5,24,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	4	↓	↓	↓	↓	↓
R9	2A to 2J	21,2,3,22, 25,5,24,4, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	27	8	↓	↓	↓	↓
R10	2B to 2J	21,2,3,22, 25,5,24,4, 27,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	7	8	↓	↓	↓	↓

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R11	2C to 2J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	6	8	40	400	ns	14,15
R12	2D to 2J	21,2,3,22, 25,5,24,4, 27,7,6,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	26					
R13	2E to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	30					
R14	2F to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	29					
R15	2G to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	9					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
R16	2H to 2J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	28	8	40	400	ns	14,15
R17	3A to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	11	13				
R18	3B to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	31					
R19	3C to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	12					
R20	3D to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,15, 34,14,33,35, 16,36,17,20, 19,38,18	32					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter		Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R21	3E to 3J	Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 34,14,33,35, 16,36,17,20, 19,38,18	15	13	40	400	ns	14,15
R22	3F to 3J		21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,14,33,35, 16,36,17,20, 19,38,18	34					
R23	3G to 3J		21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,33,35, 16,36,17,20, 19,38,18	14					
R24	3H to 3G		21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,35, 16,36,17,20, 19,38,18	33					
R25	4A to 4J		21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 16,36,17,20, 19,38,18	35	37				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R26	4B to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,36,17,20, 19,38,18	16	37	40	400	ns	14,15
R27	4C to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,17,20, 19,38,18	36					
R28	4D to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,20, 19,38,18	17					
R29	4E to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 19,38,18	20					
R30	4F to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,38,18	19					

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R31	4G to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,18	38	37	40	400	ns	14,15
R32	4H to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,38	18	37	40	400		
R33	1A to 1J Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	2,3,22,25,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	21	23	60	600		
R34	1B to 1J	21,3,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	2					
R35	1C to 1J	21,2,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	3					

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R36	1D to 1J Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,25, 5,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	22	23	60	600	ns	14,15
R37	1E to 1J	21,2,3,22, 5,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	25					
R38	1F to 1J	21,2,3,22, 25,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	5					
R39	1G to 1J	21,2,3,22, 25,5,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	24					
R40	1H to 1J	21,2,3,22, 25,5,24,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	4					

Source: https://assist.dla.mil -- Downloaded: 2026-0-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
R41	2A to 2J Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	27	8	60	600	ns	14,15
R42	2B to 2J	21,2,3,22, 25,5,24,4, 27,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	7					
R43	2C to 2J	21,2,3,22, 25,5,24,4, 27,7,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	6					
R44	2D to 2J	21,2,3,22, 25,5,24,4, 27,7,6,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	26					
R45	2E to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	30					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
R46	2F to 2J Propagation delay time (t_{PHL}), C_L = 150 pF V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	29	8	60	600	ns	14,15
R47	2G to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	9					
R48	2H to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	28					
R49	3A to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	11	13				
R50	3B to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	31	13				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R51	3C to 3J Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	12	13	60	600	ns	14,15
R52	3D to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,15, 34,14,33,35, 16,36,17,20, 19,38,18	32					
R53	3E to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 34,14,33,35, 16,36,17,20, 19,38,18	15					
R54	3F to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,14,33,35, 16,36,17,20, 19,38,18	34					
R55	3G to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,33,35, 16,36,17,20, 19,38,18	14					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R56	3H to 3G Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,35, 16,36,17,20, 19,38,18	33	13	60	600	ns	14,15
R57	4A to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 16,36,17,20, 19,38,18	35	37				
R58	4B to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,36,17,20, 19,38,18	16					
R59	4C to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,17,20, 19,38,18	36					
R60	4D to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,20, 19,38,18	17					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVII. Propagation delay time (t_{PHL}, V_{DD} = 5 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
R61	4E to 4J Propagation delay time (t_{PHL}), C_L = 150 pF, V_{DD} = 5 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 19,38,18	20	37	60	600	ns	14,15
R62	4F to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,38,18	19					
R63	4G to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,18	38					
R64	4H to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,38	18					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S1	1A to 1J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	2,3,22,25, 5,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	21	23	30	340	ns	14,15
S2	1B to 1J	21,3,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	2					
S3	1C to 1J	21,2,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	3					
S4	1D to 1J	21,2,3,25,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	22					
S5	1E to 1J	21,2,3,22, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	25					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S6	1F to 1J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	5	23	30	340	ns	14,15
S7	1G to 1J	21,2,3,22, 25,5,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	24					
S8	1H to 1J	21,2,3,22, 25,5,24,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	4					
S9	2A to 2J	21,2,3,22, 25,5,24,4, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	27	8				
S10	2B to 2J	21,2,3,22, 25,5,24,4, 27,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	7	8				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S11	2C to 2J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	6	8	30	340	ns	14,15
S12	2D to 2J	21,2,3,22, 25,5,24,4, 27,7,6,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	26					
S13	2E to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	30					
S14	2F to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	29					
S15	2G to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	9					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S16	2H to 2J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	28	8	30	340	ns	14,15
S17	3A to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	11	13				
S18	3B to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	31					
S19	3C to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	12					
S20	3D to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,15, 34,14,33,35, 16,36,17,20, 19,38,18	32					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S21	3E to 3J Propagation delay time (t_{PLH}), C_L = 50 pF; V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 34,14,33,35, 16,36,17,20, 19,38,18	15	13	30	340	ns	14,15
S22	3F to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,14,33,35, 16,36,17,20, 19,38,18	34					
S23	3G to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,33,35, 16,36,17,20, 19,38,18	14					
S24	3H to 3G	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,35, 16,36,17,20, 19,38,18	33					
S25	4A to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 16,36,17,20, 19,38,18	35	37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S26	4B to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,36,17,20, 19,38,18	16	37	30	340	ns	14,15
S27	4C to 4J	21,2,3,22 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,17,20, 19,38,18	36					
S28	4D to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,20, 19,38,18	17					
S29	4E to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 19,38,18	20					
S30	4F to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,38,18	19					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Propagation delay time (t_{PLH}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
S31	4G to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,18	38	37	30	340	ns	14,15
S32	4H to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,38	18	37	30	340	ns	14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIX. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T1	1A to 1J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	2,3,22,25, 5,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	21	23	15	160	ns	14,15
T2	1B to 1J	21,3,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	2					
T3	1C to 1J	21,2,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	3					
T4	1D to 1J	21,2,3,25,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	22					
T5	1E to 1J	21,2,3,22,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	25					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIX. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T6	1F to 1J Propagation delay time, (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	5	23	15	160	ns	14,15
T7	1G to 1J	21,2,3,22, 25,5,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	24					
T8	1H to 1J	21,2,3,22, 25,5,24,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	4					
T9	2A to 2J	21,2,3,22, 25,5,24,4,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	27	8				
T10	2B to 2J	21,2,3,22, 25,5,24,4, 27,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	7	8				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIX. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T11	2C to 2J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	6	8	15	160	ns	14,15
T12	2D to 2J	21,2,3,22, 25,5,24,4, 27,7,6,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	26					
T13	2E to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	30					
T14	2F to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	29					
T15	2G to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	9					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIX. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T16	2H to 2J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	28	8	15	160	ns	14,15
T17	3A to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	11	13				
T18	3B to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	31					
T19	3C to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	12					
T20	3D to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,15, 34,14,33,35, 16,36,17,20, 19,38,18	32					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIX. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T21	3E to 3J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 34,14,33,35, 16,36,17,20, 19,38,18	15	13	15	160	ns	14,15
T22	3F to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,14,33,35, 16,36,17,20, 19,38,18	34					
T23	3G to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,33,35, 16,36,17,20, 19,38,18	14					
T24	3H to 3G	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,35, 16,36,17,20, 19,38,18	33					
T25	4A to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 16,36,17,20, 19,38,18	35	37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIX. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
T26	4B to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,36,17,20, 19,38,18	16	37	15	160	ns	14,15
T27	4C to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,17,20, 19,38,18	36					
T28	4D to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,20, 19,38,18	17					
T29	4E to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 19,38,18	20					
T30	4F to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,38,18	19					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XIX. Propagation delay time (t_{PHL}, V_{DD} = 10 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
T31	4G to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,18	38	37	15	160	ns	14,15
T32	4H to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 10 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,38	18	37	15	160	ns	14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U1	1A to 1J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	2,3,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	21	23	24	240	ns	14,15
U2	1B to 1J	21,3,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	2					
U3	1C to 1J	21,2,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	3					
U4	1D to 1J	21,2,3,25,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	22					
U5	1E to 1J	21,2,3,22,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	25					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
U6	1F to 1J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	5	23	24	240	ns	14,15
U7	1G to 1J	21,2,3,22, 25,5,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	24					
U8	1H to 1J	21,2,3,22, 25,5,24,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	4					
U9	2A to 2J	21,2,3,22, 25,5,24,4, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	27	8				
U10	2B to 2J	21,2,3,22, 25,5,24,4, 27,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	7	8				

Source: https://assist.dla.mil - Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U11	2C to 2J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	6	8	24	240	ns	14,15
U12	2D to 2J	21,2,3,22, 25,5,24,4, 27,7,6,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	26					
U13	2E to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	30					
U14	2F to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	29					
U15	2G to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	9					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U16	2H to 2J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	28	8	24	240	ns	14,15
U17	3A to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	11	13				
U18	3B to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	31					
U19	3C to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	12					
U20	3D to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,15, 34,14,33,35, 16,36,17,20, 19,38,18	32					

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U21	3E to 3J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 34,14,33,35, 16,36,17,20, 19,38,18	15	13	24	240	ns	14,15
U22	3F to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,14,33,35, 16,36,17,20, 19,38,18	34					
U23	3G to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,33,35, 16,36,17,20, 19,38,18	14					
U24	3H to 3G	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,35, 16,36,17,20, 19,38,18	33					
U25	4A to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 16,36,17,20, 19,38,18	35	37				

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U26	4B to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,36,17,20, 19,38,18	16	37	24	240	ns	14,15
U27	4C to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,17,20, 19,38,18	36					
U28	4D to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,20, 19,38,18	17					
U29	4E to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 19,38,18	20					
U30	4F to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,38,18	19					

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XX. Propagation delay time (t_{PLH}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
U31	4G to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,18	38	37	24	240	ns	14,15
U32	4H to 4J Propagation delay time (t_{PLH}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,38	18	37	24	240	ns	14,15

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V1	1A to 1J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	2,3,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	21	23	12	120	ns	14,15
V2	1B to 1J	21,3,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	2					
V3	1C to 1J	21,2,22,25, 5,24,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	3					
V4	1D to 1J	21,2,3,25,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	22					
V5	1E to 1J	21,2,3,22,5, 24,4,27,7,6, 26,30,29,9, 28,11,31,12, 32,15,34,14, 33,35,16,36, 17,20,19,38, 18	25					

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin.	Min	Max		
V6	1F to 1J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,24,4,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	5	23	12	120	ns	14,15
V7	1G to 1J	21,2,3,22, 25,5,4,27,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	24					
V8	1H to 1J	21,2,3,22, 25,5,24,27, 7,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	4					
V9	2A to 2J	21,2,3,22, 25,5,24,4,7, 6,26,30,29, 9,28,11,31, 12,32,15,34, 14,33,35,16, 36,17,20,19, 38,18	27	8				
V10	2B to 2J	21,2,3,22, 25,5,24,4, 27,6,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	7	8				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V11	2C to 2J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,26,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	6	8	12	120	ns	14,15
V12	2D to 2J	21,2,3,22, 25,5,24,4, 27,7,6,30, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	26					
V13	2E to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 29,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	30					
V14	2F to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,9,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	29					
V15	2G to 2J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,28,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	9					

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V16	2H to 2J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,11, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	28	8	12	120	ns	14,15
V17	3A to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 31,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	11	13				
V18	3B to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,12,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	31					
V19	3C to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,32,15, 34,14,33,35, 16,36,17,20, 19,38,18	12					
V20	3D to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,15, 34,14,33,35, 16,36,17,20, 19,38,18	32					

Source: https://assist.dla.mil - Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25° ±5°C ambient Min	Initial limits 25° ±5°C ambient Max	Unit	Test figure
V21	3E to 3J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 34,14,33,35, 16,36,17,20, 19,38,18	15	13	12	120	ns	14,15
V22	3F to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,14,33,35, 16,36,17,20, 19,38,18	34					
V23	3G to 3J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,33,35, 16,36,17,20, 19,38,18	14					
V24	3H to 3G	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,35, 16,36,17,20, 19,38,18	33					
V25	4A to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 16,36,17,20, 19,38,18	35	37				

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V26	4B to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,36,17,20, 19,38,18	16	37	12	120	ns	14,15
V27	4C to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,17,20, 19,38,18	36					
V28	4D to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,20, 19,38,18	17					
V29	4E to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 19,38,18	20					
V30	4F to 4J	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,38,18	19					

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XXI. Propagation delay time (t_{PHL}, V_{DD} = 15 V) test - Continued.

Test condition	Parameter	Connect			Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
V31	4G to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,18	38	37	12	120	ns	14,15
V32	4H to 4J Propagation delay time (t_{PHL}), C_L = 50 pF, V_{DD} = 15 V	21,2,3,22, 25,5,24,4, 27,7,6,26, 30,29,9,28, 11,31,12,32, 15,34,14,33, 35,16,36,17, 20,19,38	18	37	12	120	ns	14,15

TABLE XXII. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
W	Power supply filter capacitance	1	Open	0.007	0.013	µF	16

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Input capacitance test.

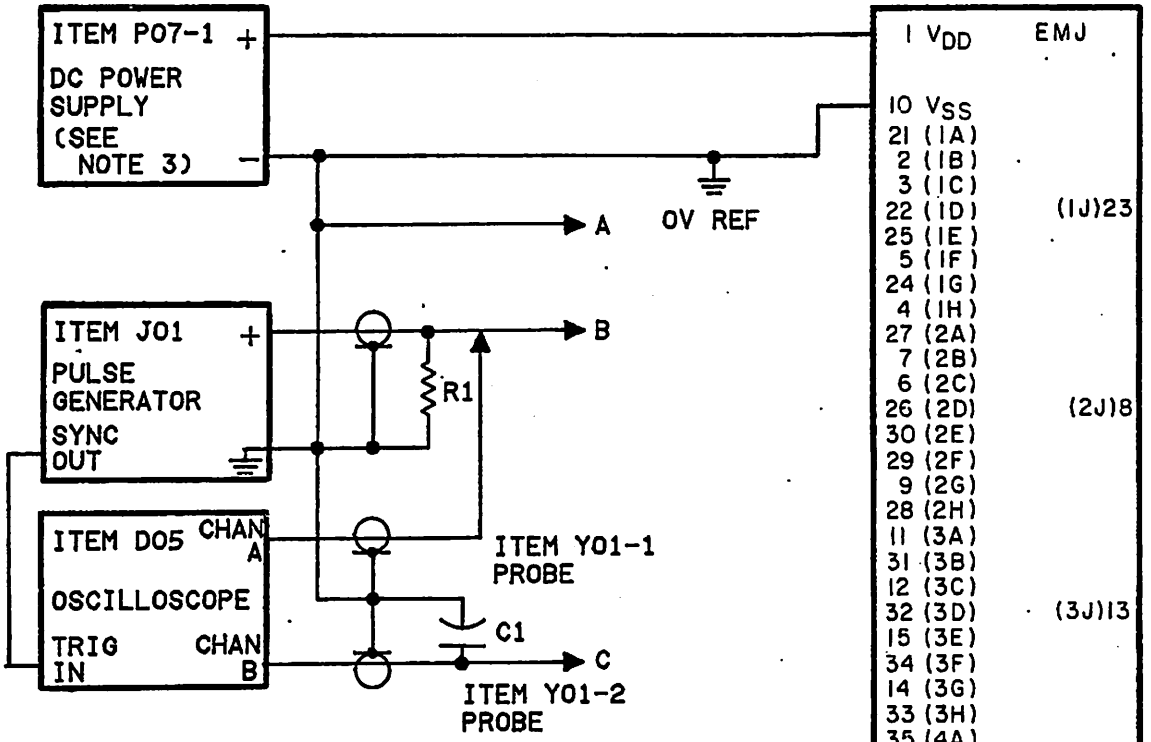
Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X1	1A Input capacitance	Open	21	2	15	pF	16
X2	1B		2				
X3	1C		3				
X4	1D		22				
X5	1E		25				
X6	1F		5				
X7	1G		24				
X8	1H		4				
X9	2A		27				
X10	2B		7				
X11	2C		6				
X12	2D		26				
X13	2E		30				
X14	2F		29				
X15	2G		9				
X16	2H		28				
X17	3A		11				
X18	3B		31				
X19	3C		12				
X20	3D		32				

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T17:10Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25° ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X21	3E Input capacitance	Open	15	2	15	pF	16
X22	3F		34				
X23	3G		14				
X24	3H		33				
X25	4A		35				
X26	4B		16				
X27	4C		36				
X28	4D		17				
X29	4E		20				
X30	4F		19				
X31	4G		38				
X32	4H		18				

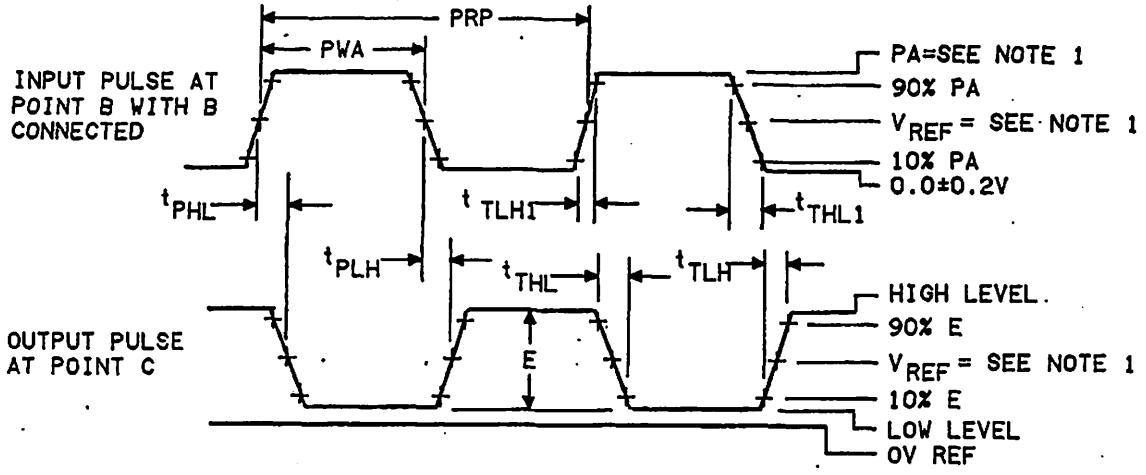
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



NOTES: 1. R1 = 51.1 Ω ±5%, 5 W.
2. C1 = 50 ±5 pF for test conditions Q1-Q32, R1-R32, S1-S32, T1-T32, U1-U32 and V1-V32 including probe and parasitic capacitance.
= 150 ±15 pF for test conditions Q33-Q64 and R33-R64 including probe and parasitic capacitance.
3. Item P07-1 Voltage:
= 5.0 V ±1% for test conditions Q1-Q64 and R1-R64.
= 10.0 V ±1% for test conditions S1-S32 and T1-T32.
= 15.0 V ±1% for test conditions U1-U32 and V1-V32.

FIGURE 14. Propagation delay time (t_{PLH} and t_{PHL}) test circuit for tables XVI through XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



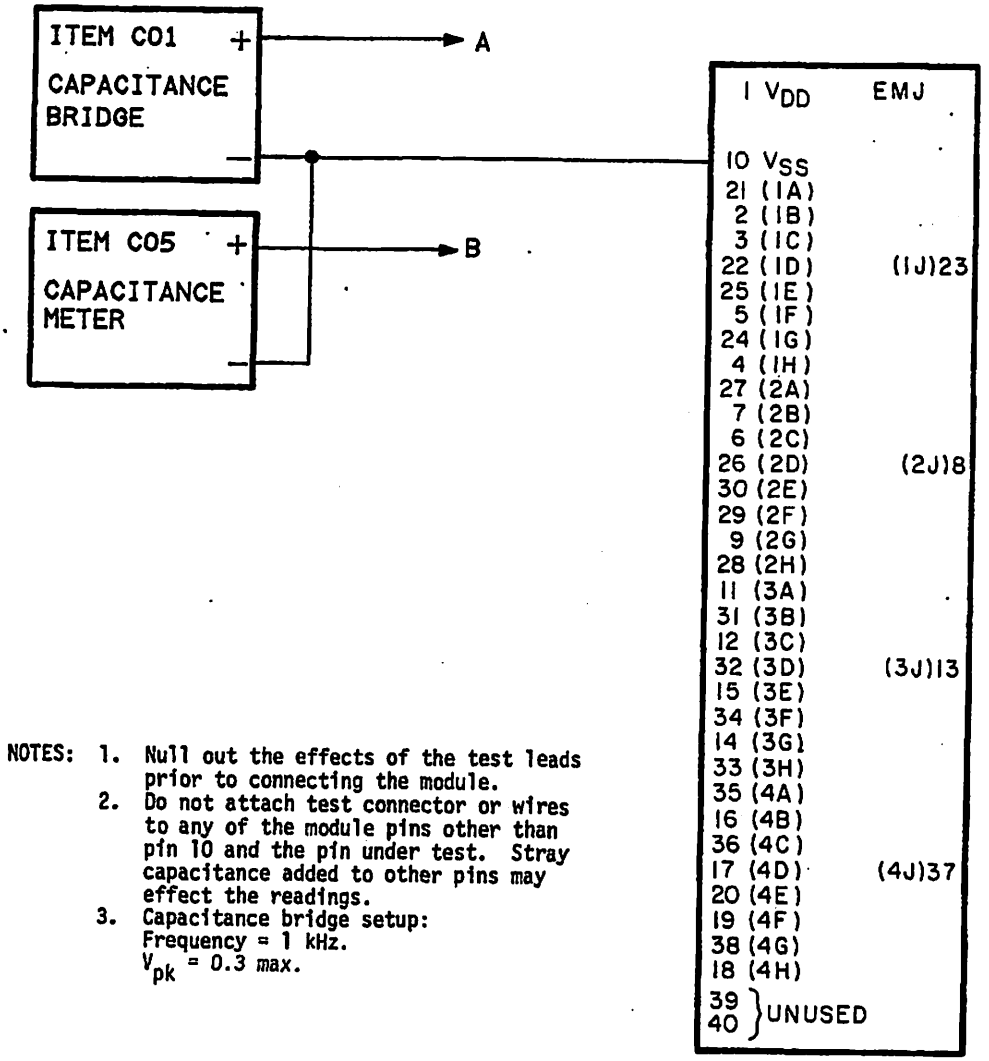
NOTES: 1. Input pulse parameters:
PWA = 2.5 ±0.2 µs for test conditions L1-L8, M1-M8, Q1-Q64 and R1-R64.
= 1.0 ±0.1 µs for test conditions N1-N8, P1-P8, S1-S32, T1-T32, U1-U32 and V1-V32.
RRP = 5.0 ±0.5 µs.
$t_{TLH1} = t_{THL1}$ = 10 ±2 ns.
PA = 5.0 ±0.1 V for test conditions L1-L8, M1-M8, Q1-Q64 and R1-R64.
= 10.0 ±0.1 V for test conditions N1-N8, S1-S32 and T1-T32.
= 15.0 ±0.1 V for test conditions P1-P8, U1-U32 and V1-V32.
V_{REF} = 2.5 ±0.1 V for test conditions L1-L8, M1-M8, Q1-Q64 and R1-R64.
= 5.0 ±0.1 V for test conditions N1-N8, S1-S32 and T1-T32.
= 7.5 ±0.1 V for test conditions P1-P8, U1-U32 and V1-V32.

2. Before making the measurements, verify that the output pulse level is:
Output pulse high level ≥ 4.6 V for test conditions L1-L8, M1-M8, Q1-Q64 and R1-R64.
≥ 9.5 V for test conditions N1-N8, S1-S32 and T1-T32.
≥ 13.5 V for test conditions P1-P8, U1-U32 and V1-V32.
Output pulse low level ≤ 0.4 V for test conditions L1-L8, M1-M8, Q1-Q64 and R1-R64.
≤ 0.5 V for test conditions N1-N8, S1-S32 and T1-T32.
≤ 1.5 V for test conditions P1-P8, U1-U32 and V1-V32.

3. The lead lengths of electrical conductors between resistor R1 and input pins, probes YO1-1 and YO1-2 to input and output pins and between load circuit and output pin shall be less than one inch (25.4 mm).
4. All voltages are referenced to V_{SS}.

FIGURE 15: Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for tables XII through XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:10Z
Check the source to verify that this is the current version before use.



NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may effect the readings.
3. Capacitance bridge setup:
   Frequency = 1 kHz.
   V_{pk} = 0.3 max.

FIGURE 16. Power supply filter and input capacitance test circuit for tables XXII and XXIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T17:10Z
Check the source to verify that this is the current version before use.